

Homework 3

Caroline Hutchings

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Homework 3 - Tidyverse

Conceptual Questions

1. If your working directory is `myfolder/homework/`, what relative path would you specify to get the file located at `myfolder/MyData.csv`?
 1. `"../MyData.csv"`
2. What are the major benefits of using R projects?
 1. I can create a project organization directly in RStudio, set the working directory, and manage git and github directly in RStudio (I love this!).
3. What is Git and what is GitHub?
 1. Git is a version control software that is run locally on my computer. It tracks changes to files and allows me to make those changes permanent or revert back to a previous version.
 2. GitHub is a cloud-based platform that hosts git repositories. It is a remote backup of repositories. It allows collaboration between team members by tracking changes and allowing each team member to upload their changes and others to pull those changes to their local environment. It also tracks versions and will alert the team member if there are merge conflicts allowing the team member to decide how to handle the conflicts.
4. What are the two main differences between a tibble and a `data.frame`?
 1. If one column is called from a tibble the tibble allows the column to remain a tibble instead of a vector.
 2. The tibble displays the data in a nicer format by only printing the data that is visible in the output. It also lists the number of columns and rows along with all the column names and types.
 3. Tibbles do not automatically convert character variables to factors.

Reading Delimited Data

Glass Data

Locate the glass data and read the comma separated values files into a tibble using the tidyverse function `read_csv`, but first load the library to load the functions in this session.

```
library(readr)
#?read_csv
glass_data <- read_csv("https://www4.stat.ncsu.edu/online/datasets/glass.data", col_names=c(
```

Rows: 214 Columns: 11

-- Column specification -----

Delimiter: ","

dbl (11): Id, RI, Na, Mg, Al, Si, K, Ca, Ba, Fe, Type

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
glass_data
```

A tibble: 214 x 11

	Id	RI	Na	Mg	Al	Si	K	Ca	Ba	Fe	Type
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	1.52	13.6	4.49	1.1	71.8	0.06	8.75	0	0	1
2	2	1.52	13.9	3.6	1.36	72.7	0.48	7.83	0	0	1
3	3	1.52	13.5	3.55	1.54	73.0	0.39	7.78	0	0	1
4	4	1.52	13.2	3.69	1.29	72.6	0.57	8.22	0	0	1
5	5	1.52	13.3	3.62	1.24	73.1	0.55	8.07	0	0	1
6	6	1.52	12.8	3.61	1.62	73.0	0.64	8.07	0	0.26	1
7	7	1.52	13.3	3.6	1.14	73.1	0.58	8.17	0	0	1
8	8	1.52	13.2	3.61	1.05	73.2	0.57	8.24	0	0	1
9	9	1.52	14.0	3.58	1.37	72.1	0.56	8.3	0	0	1
10	10	1.52	13	3.6	1.36	73.0	0.57	8.4	0	0.11	1

i 204 more rows

Yeast Data

Identify the correct number spaces for the start of each column to read the fixed width data into a tibble using the `read_delim` function since the data is fixed width not csv. Count the

spaces for the start of each column ADT1_YEAST 0.58 0.61 0.47 0.13 0.50 0.00 0.48 0.22
MIT

```
#?read_fwf
yeast_data <- read_fwf('https://www4.stat.ncsu.edu/online/datasets/yeast.data', fwf_widths(c
```

Rows: 1484 Columns: 10

-- Column specification -----

chr (2): Sequence, class

dbl (8): mcg, gvh, alm, mit, erl, pox, vac, nuc

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
yeast_data
```

A tibble: 1,484 x 10

	Sequence <chr>	mcg <dbl>	gvh <dbl>	alm <dbl>	mit <dbl>	erl <dbl>	pox <dbl>	vac <dbl>	nuc <dbl>	class <chr>
1	ADT1_YEAST	0.58	0.61	0.47	0.13	0.5	0	0.48	0.22	MIT
2	ADT2_YEAST	0.43	0.67	0.48	0.27	0.5	0	0.53	0.22	MIT
3	ADT3_YEAST	0.64	0.62	0.49	0.15	0.5	0	0.53	0.22	MIT
4	AAR2_YEAST	0.58	0.44	0.57	0.13	0.5	0	0.54	0.22	NUC
5	AATM_YEAST	0.42	0.44	0.48	0.54	0.5	0	0.48	0.22	MIT
6	AATC_YEAST	0.51	0.4	0.56	0.17	0.5	0.5	0.49	0.22	CYT
7	ABC1_YEAST	0.5	0.54	0.48	0.65	0.5	0	0.53	0.22	MIT
8	BAF1_YEAST	0.48	0.45	0.59	0.2	0.5	0	0.58	0.34	NUC
9	ABF2_YEAST	0.55	0.5	0.66	0.36	0.5	0	0.49	0.22	MIT
10	ABP1_YEAST	0.4	0.39	0.6	0.15	0.5	0	0.58	0.3	CYT

i 1,474 more rows